

JKBOSE 2026

CLASS 11TH

Photosynthesis in Higher Plants

LECTURE-04

BOTANY

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Topics To Covered

- 1** ABSORPTION SPECTRUM: (T.W ENGELMANN)
- 2** PIGMENT SYSTEM OR PHOTOSYSTEM



**JKBOSE
WALLAH**

**THE SESSION THAT
CAN *CHANGE*
YOUR BOARD PREPARATION.**



➤ **EXCLUSIVE ONLINE SEMINAR** ⚡
ONLY FOR **PAID BATCH STUDENTS**

**ROADMAP
TO BOARDS**

- 📖 What to study first
- 📈 How to score higher in Boards
- 🎯 Smart preparation strategy
- ⚠️ Common mistakes to avoid
- 🕒 Time management & revision plan
- ✅ Complete plan till Board Exams

ONE SESSION. COMPLETE CLARITY.



TOMORROW'S SCHEDULE ⚡

CLASS	TIME
🎓 9 th	1:00 PM
🎓 10 th	7:30 PM
🎓 11 th	9:00 AM
🎓 12 th	5:00 PM
🎓 11 th (HUMANITIES)	8:00 PM
🎓 12 th (HUMANITIES)	6:00 PM



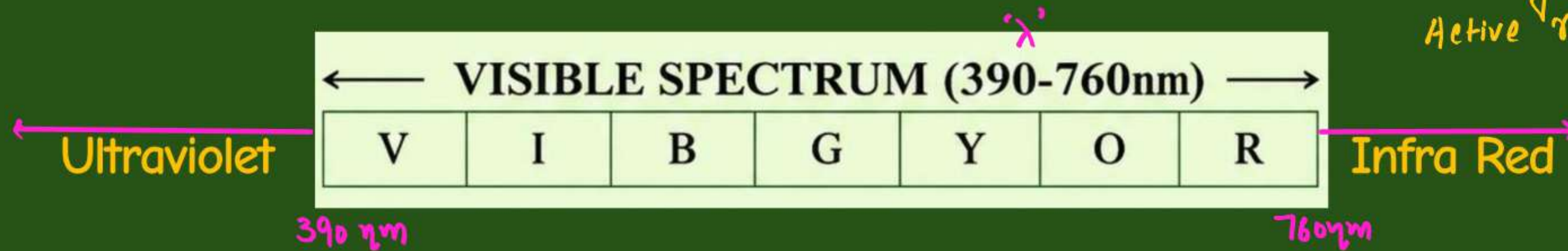
BE THERE. DON'T MISS IT! ⚡



**IMPORTANT SESSION FOR
YOUR BOARD SUCCESS** ⚡

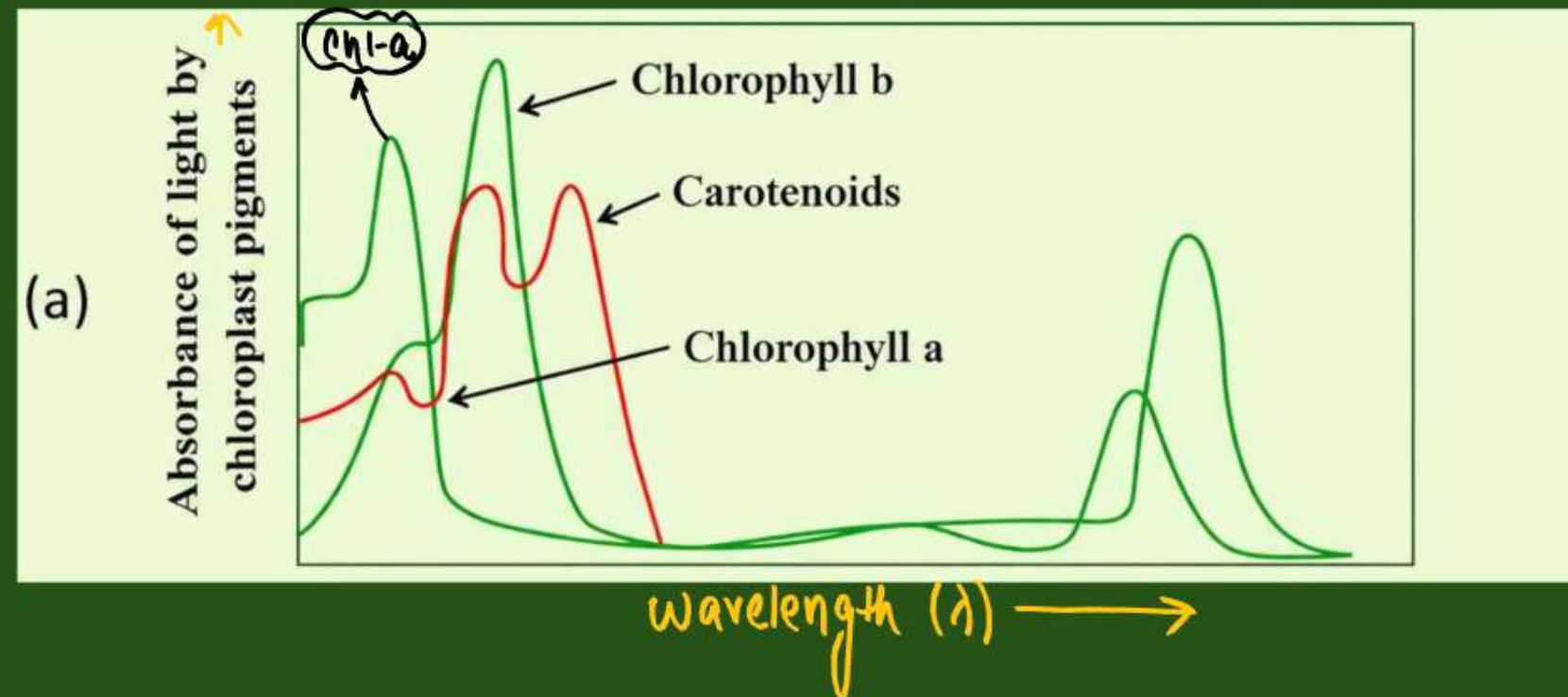
ABSORPTION SPECTRUM: (T.W ENGELMANN)

* PAR $\downarrow$
Photosynthetically
Active radiations (400-700nm)



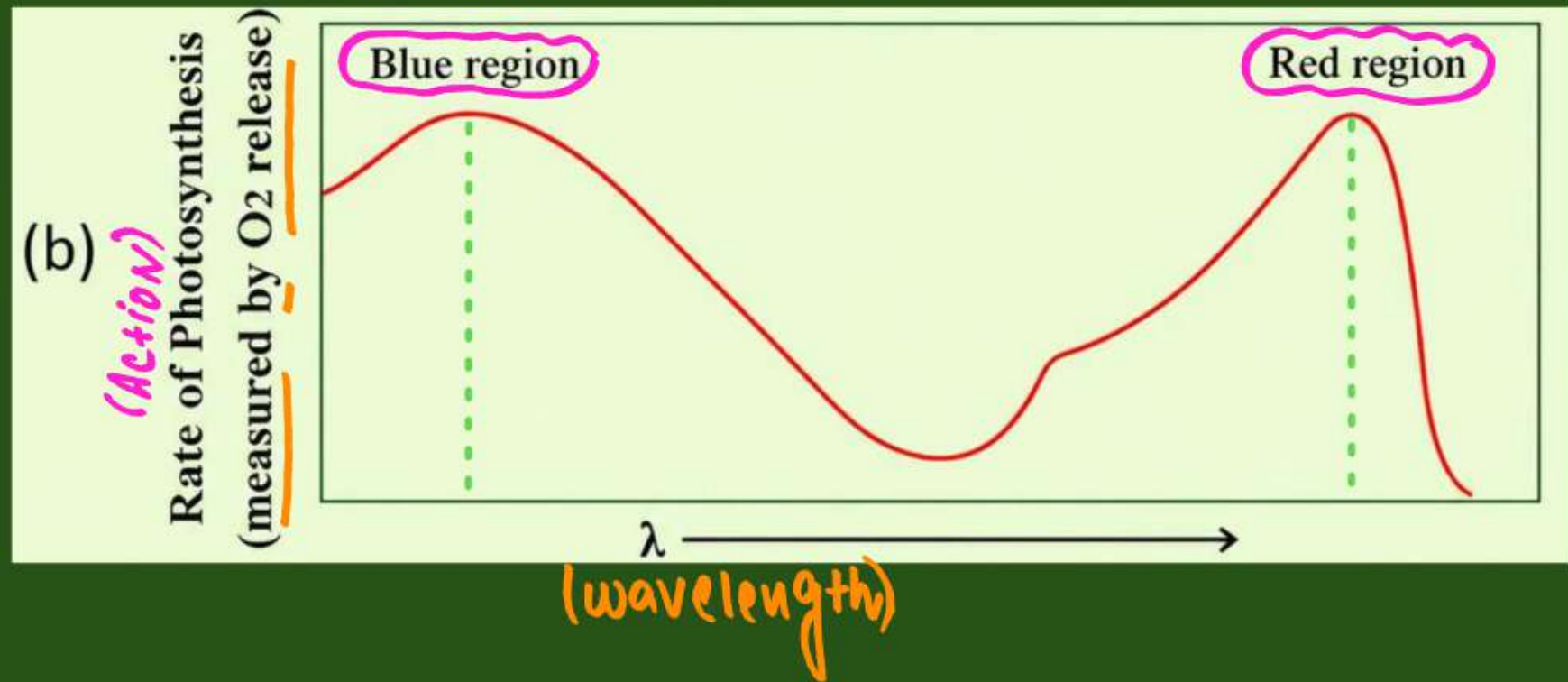
Absorption Spectrum:

- It is a curve/graph which is obtained by absorption of different wavelengths by different photosynthetic pigments.
- It gives absorption maxima and absorption minima of a pigment.
- It is also called as finger-print of a photosynthetic pigment.



Action Spectrum → T.W Englemann.

↓
Already discussed.



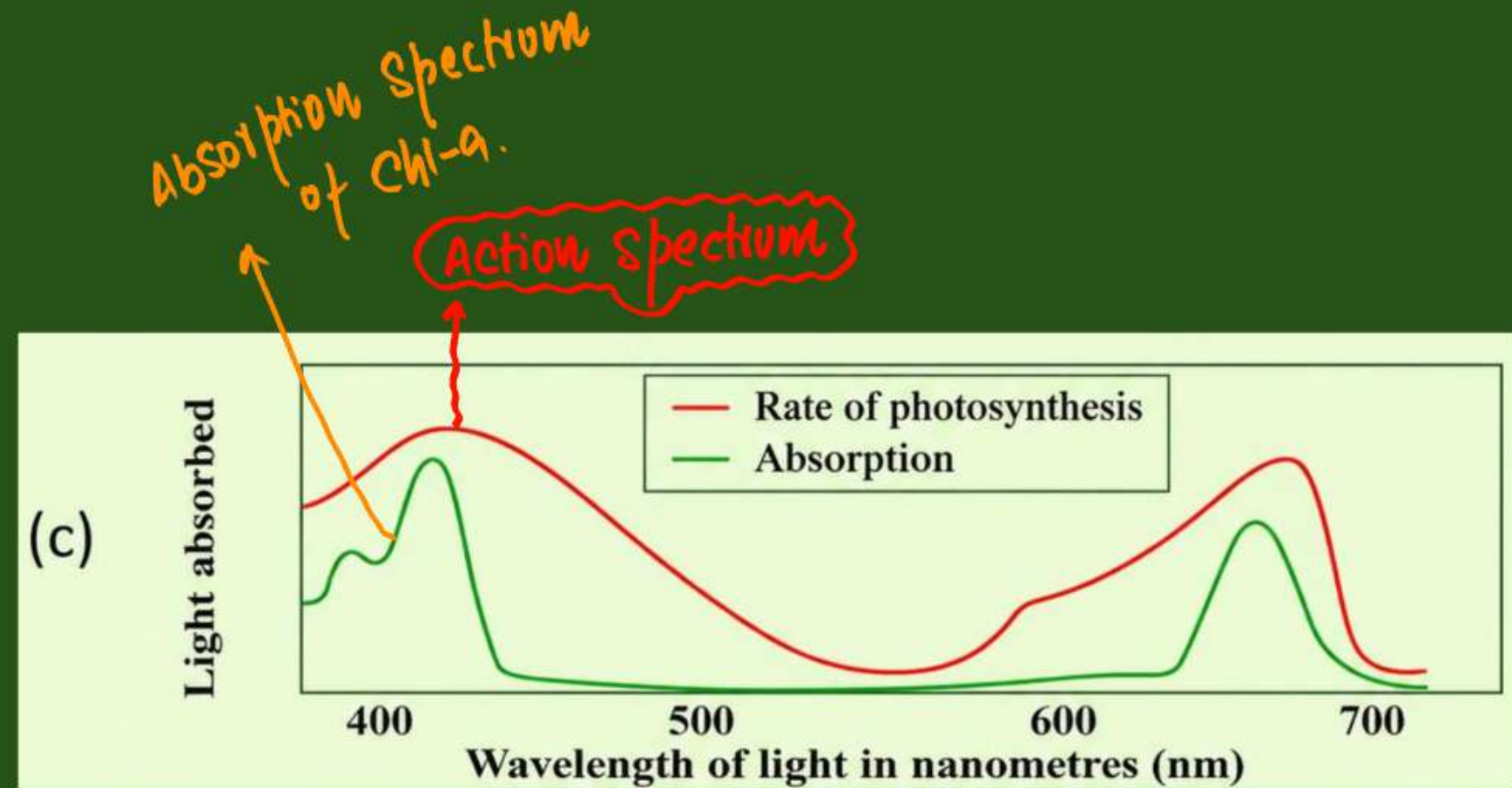
NOTE overlapping of absorption spectrum of chl-a and Action Spectrum.



≡ There is close match b/w absorption spectrum of chl-a and Action spectrum.

≡ But one-to-one overlap is not there.

Conclusion chl-a is chief photosynthetic pigment.

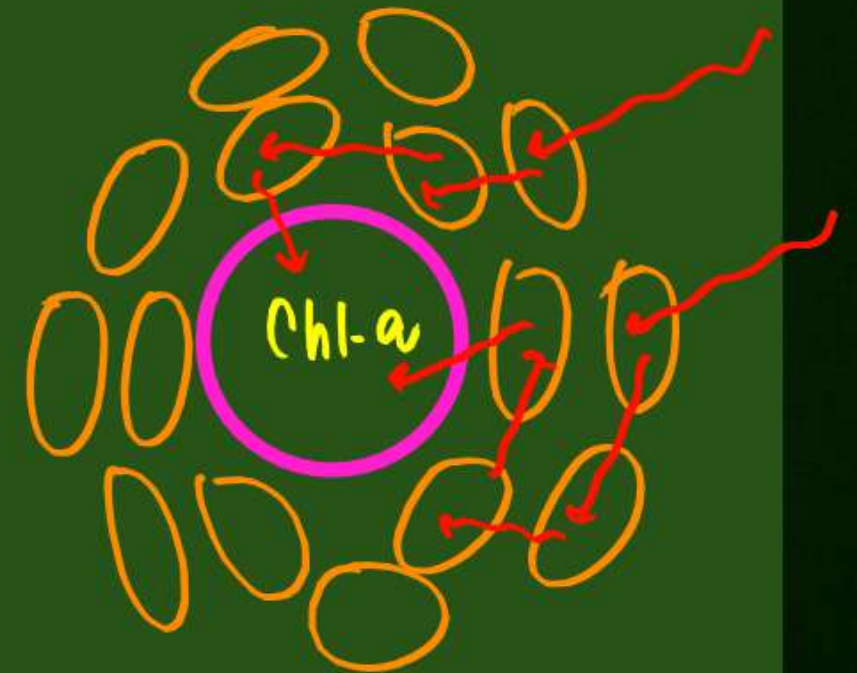


Important points

- ① Maximum photosynthesis occurs in blue and red region of Visible Spectrum.
- ② Some photosynthesis also takes place at other wavelengths of Visible Spectrum.
- ③ **Chl-b, Carotene, Xanthophyll** → are accessory photosynthetic pigments.

↳ They absorb wide range of wavelengths and then handover it to chl-a. Thus makes the photosynthesis more efficient.

↳ They also prevent Solarisation / photooxidation (i.e., degradation of chlorophyll) by high light intensity.



PIGMENT SYSTEM OR PHOTOSYSTEM

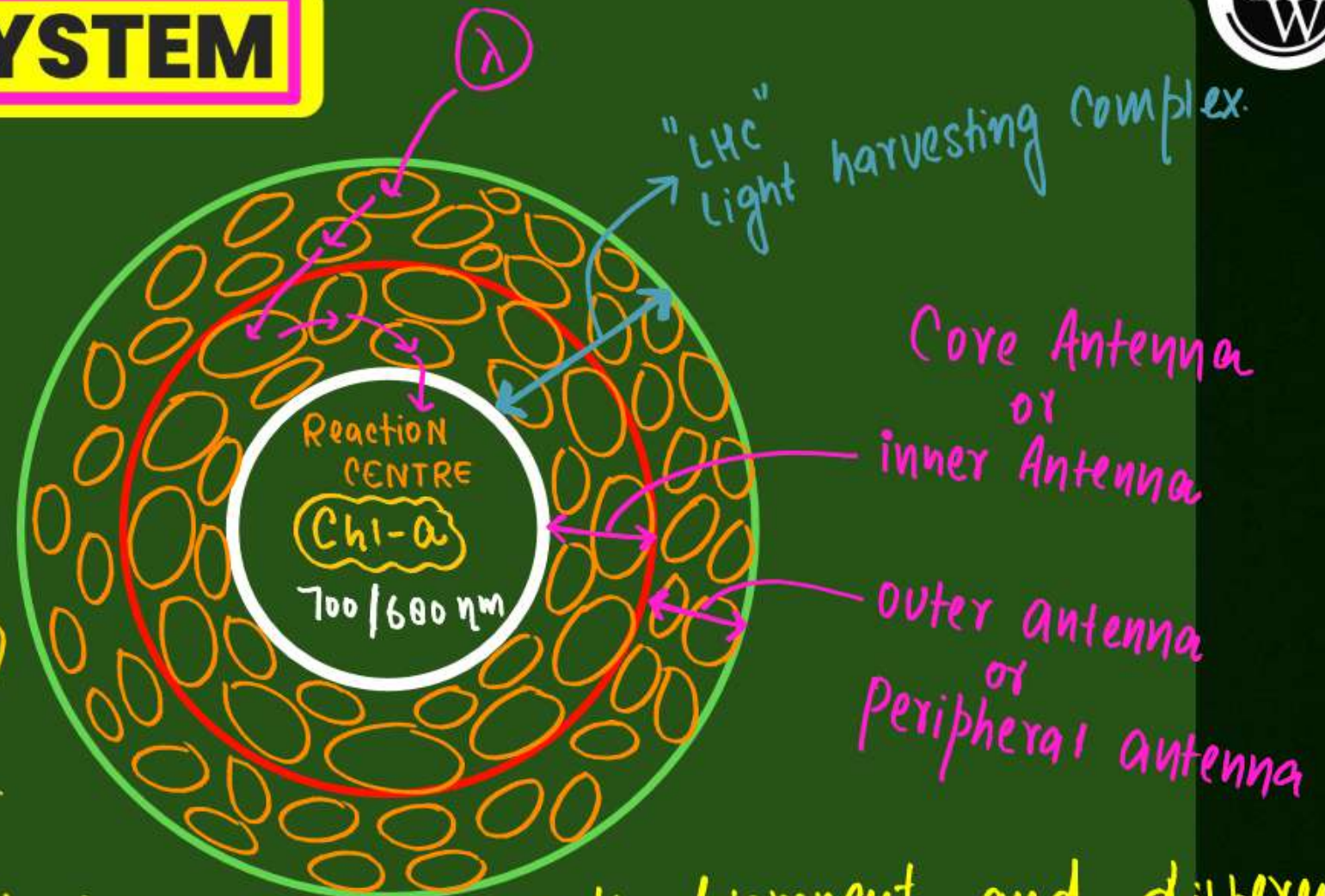
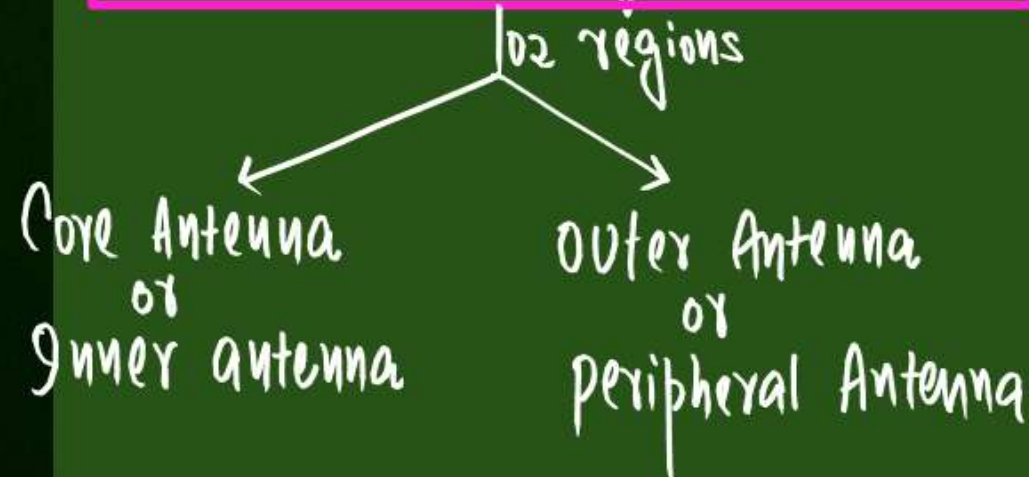
- Previously called as Quatasome.
- Placed on thylakoid membranes.
- Pigment System / photosystem consists of:

a. Reaction Centre

- Composed of single chl-a mol.
- Performs primary reactions of photosynthesis (i.e., ATP and NADPH+H⁺ formation).

b. Light Harvesting Complex (LHC)

- It is composed of accessory photosynthetic pigment and different hundreds of proteins.
- It absorbs light of different wavelengths and transfer it to Oxm. Centre.



• Photosystem is of Two types :

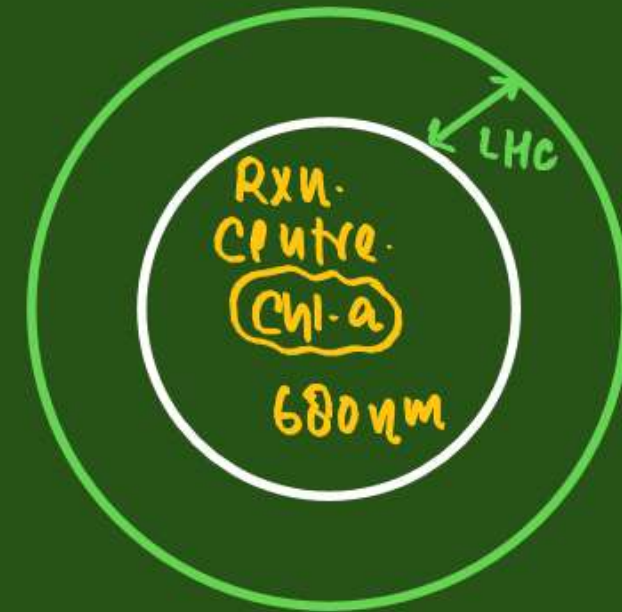
PS-I / P₇₀₀ (Photosystem-I)

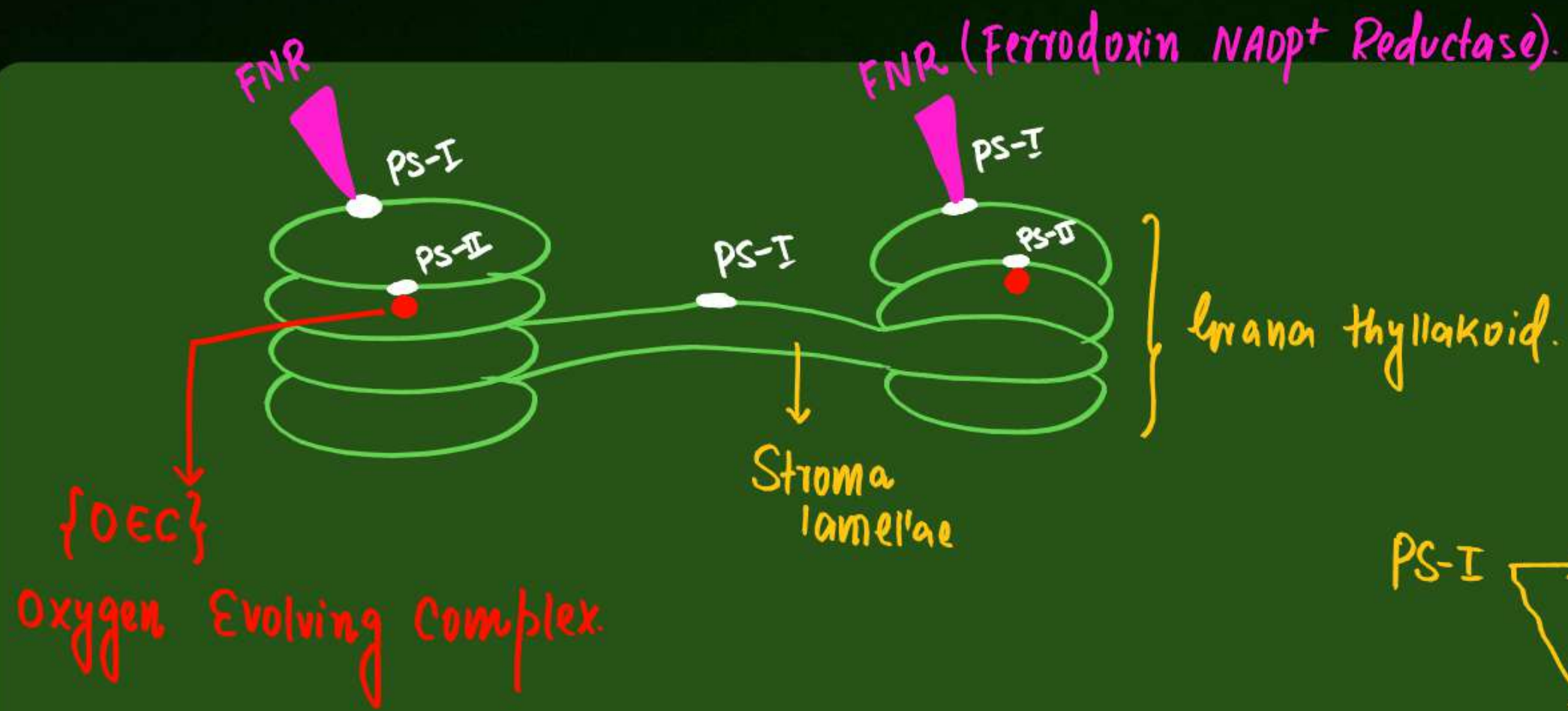
↓
discovered first.



PS-II / P₆₈₀ (Photosystem-II)

↓
discovered later i.e,
after PS-I (P₇₀₀).





PS-I → present in exposed/non-appressed membranes of thylakoid
 present in both-Grana thylakoid and Stroma lamellae.

PS-II → present in non-exposed or appressed membranes of thylakoid.
 present only in Grana thylakoid.

QUESTIONS

The reaction centre in PS II has an absorption maxima at

(2023)

A 780 nm

B 680 nm

C 700 nm

D 600 nm

QUESTIONS

H/W

Which out of the following statements is incorrect?

(2023-Manipur)

- A** Grana lamellae have both PS I and PS II
- B** Cyclic photophosphorylation involves both PS I and PS II
- C** Both ATP and NADPH + H^+ are synthesised during non-cyclic photophosphorylation
- D** Stroma lamellae lack PS II and NADP reductase

QUESTIONS

H/v

Photosynthetic active radiation (PAR) in photosynthesis lies in between

- A** 100 - 40nm
- B** 400 - 700nm
- C** 760 - 950nm
- D** 10 - 100nm

u/w

QUESTIONS

Assertion: Presence of accessory pigments enhances rate of photosynthesis.

Reason: They enable a wider range of wavelength of incoming light to be utilised for photosynthesis.

- A** If both assertion and reason are true and reason is the correct explanation of assertion.
- B** If both assertion and reason are true but reason is not the correct explanation of assertion.
- C** If assertion is true but reason is false.
- D** If both assertion and reason are false.

HOMework

3 Questions
+ Revision





Thank
You